Packet Tracer - WAN Concepts

Thitipong Karnjanadamrongnan - 633020290-2

Topology Overview

The topology contains three consumer WAN technologies (Home Cable, Home DSL and Cellular) and one Business WAN (Business Headquarters with three stores), all connected through the ISP/Telco cloud to a web server (209.165.200.226).

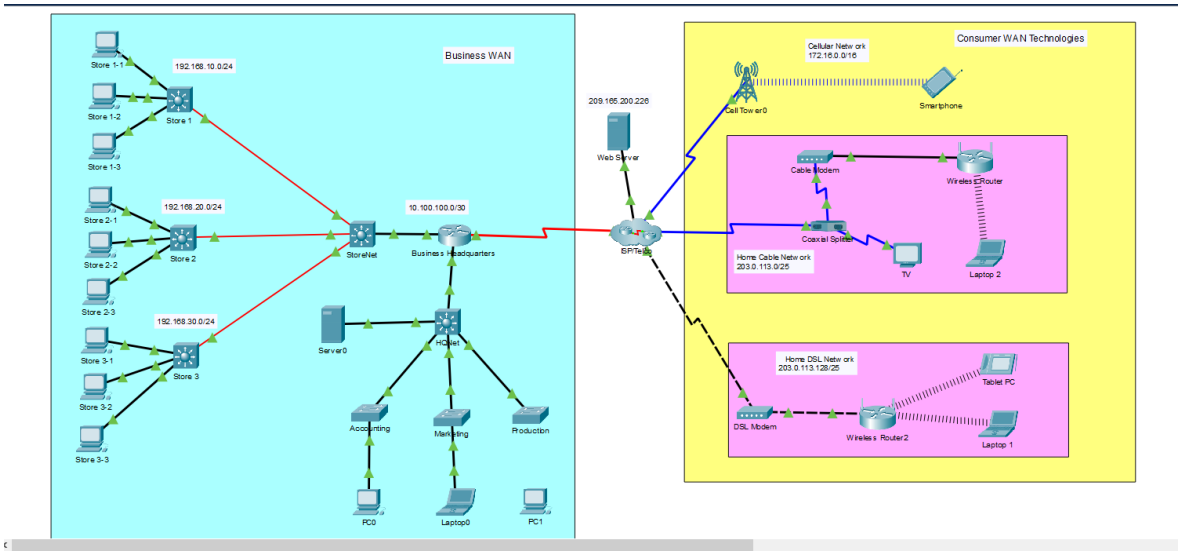


Figure 1. Network topology

Part 1: Investigate Consumer WAN Technologies

Step 1: Explore Consumer WAN Technologies

a. What are the WAN technologies in use?

The Home Cable Network (203.0.113.0/25) uses **cable** and the Home DSL Network (203.0.113.128/25) uses **DSL**. The smartphone uses a third technology, **cellular (3G/4G)**, through Cell Tower0.

b. What media is used to connect the two home networks to the ISP? What devices are directly connected to the ISP?

DSL uses a **telephone line (copper)**, and cable uses **coaxial cable**. The devices directly connected to the ISP are the **DSL modem** (DSL network) and the **coaxial splitter** (cable network).

c. DSL modem: ports, purpose, and the connection type to the cable network (why is the splitter necessary?)

The DSL modem has a **LINE 0** port (RJ-11, telephone line from the ISP) and a **FAST ETH 0** port (RJ-45, Ethernet to Wireless Router2). Its purpose is to convert the DSL signal on the telephone line into Ethernet for the home router. The ISP connects to the Home Cable Network with **coaxial cable**. The splitter is necessary because one coaxial line carries both TV and Internet signals, so it is split into two paths: one to the TV and one to the cable modem.

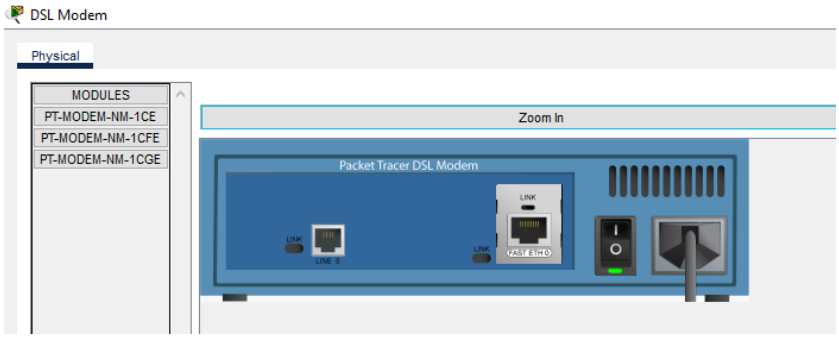


Figure 2. DSL modem - Physical tab

d. Cable modem: what does it do, what connections does it have, which router port does it connect to, and where did the IP come from?

The cable modem converts the cable (coaxial) signal into Ethernet. It has a **COAXIAL** port (from the splitter) and a **FAST ETH 0** port (to the wireless router). The cable from the modem connects to the **Internet** port (blue) of the wireless router, and the router's interface IP address is obtained from the ISP through **DHCP**.

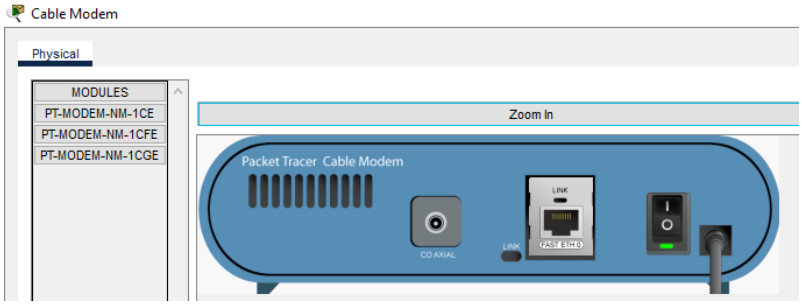


Figure 3. Cable modem - Physical tab



Figure 4. Wireless router - Physical tab (Internet port and Ethernet ports 1-4)

e. Smartphone: what is its IP address, where did it come from, and which data service is used?

The smartphone's IP address is **198.51.100.101** (mask 255.255.255.0, gateway 198.51.100.1). It is assigned by **DHCP** on the **3G/4G Cell1** interface from the cellular provider via the cell tower. The phone is using **cellular data**, not Wi-Fi.

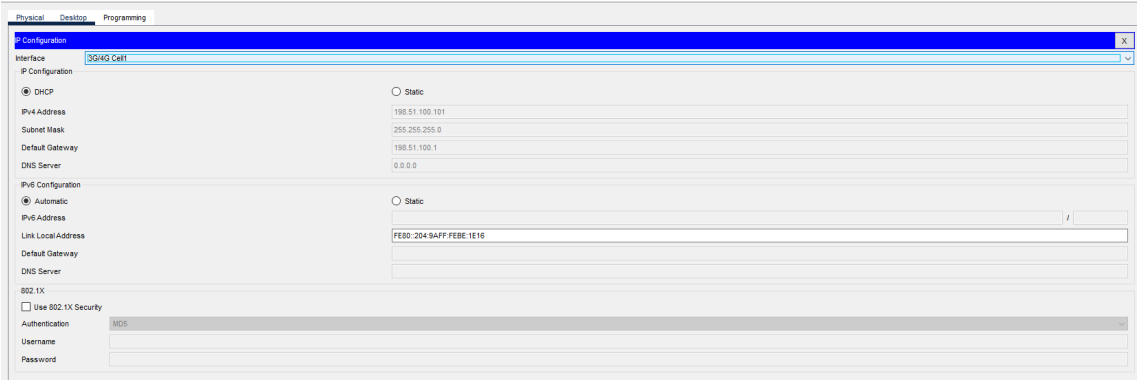


Figure 5. Smartphone - IP Configuration (3G/4G Cell1)

Step 2: Explore the Business WAN

a. What different types of connections are in use in the Business network?

Three types are used: **copper Ethernet** (black lines) inside the store and HQ LANs, **fiber-optic** (red solid lines) from the stores to the StoreNet switch, and **serial** (red zigzag line) from the Business Headquarters router to the ISP.

b. StoreNet switch: what interface types are present? Which interfaces and media connect the store networks to the Business Headquarters network, and why?

StoreNet (Catalyst 3650 24 PoE+ 2x10G) has **24 copper Gigabit Ethernet ports** (Gi1/0/1-24, all down) and **4 SFP uplink ports** (Gi1/1/1-4, all up). Gi1/1/1, Gi1/1/2 and Gi1/1/3 (10.1.1.1, 10.1.1.5, 10.1.1.9) connect to Store 1, 2 and 3 over fiber-optic cable, and Gi1/1/4 (10.100.100.2) connects to the Business Headquarters router. Fiber is used because the stores are far from headquarters: it supports longer distances and higher bandwidth than copper and is not affected by electrical interference.

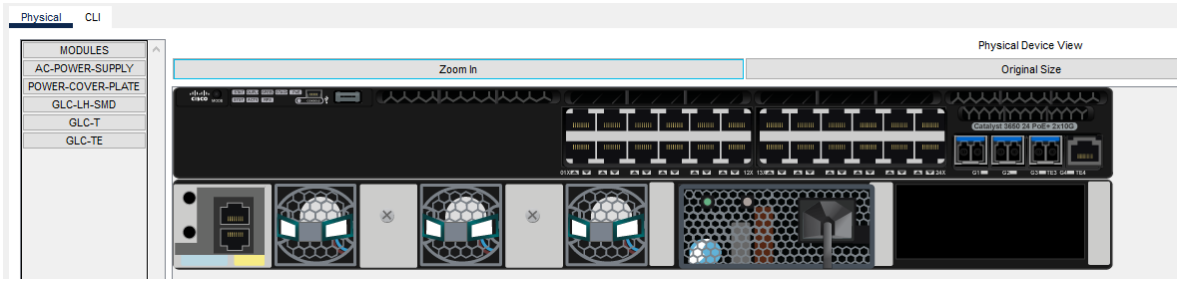


Figure 6. StoreNet - Physical view (24 copper ports and 4 SFP uplinks)

```
Switch>enable
Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/2   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/3   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/4   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/5   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/6   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/7   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/8   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/9   unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/10  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/11  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/12  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/13  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/14  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/15  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/16  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/17  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/18  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/19  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/20  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/21  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/22  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/23  unassigned      YES NVRAM  down                  down
GigabitEthernet1/0/24  unassigned      YES NVRAM  down                  down
GigabitEthernet1/1/1   10.1.1.1        YES NVRAM  up                    up
GigabitEthernet1/1/2   10.1.1.5        YES NVRAM  up                    up
GigabitEthernet1/1/3   10.1.1.9        YES NVRAM  up                    up
GigabitEthernet1/1/4   10.100.100.2    YES NVRAM  up                    up
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 unassigned      YES unset  down                  down
Switch#
```

Figure 7. StoreNet - show ip interface brief

c. What type of WAN service connects the Business Headquarters router to the ISP?

A **serial leased line** (point-to-point WAN link) leased from the Telco: Se0/1/0 on the HQ router to Se0/0/0 on the ISP. The clock icon on the ISP side shows that it is the DCE end.

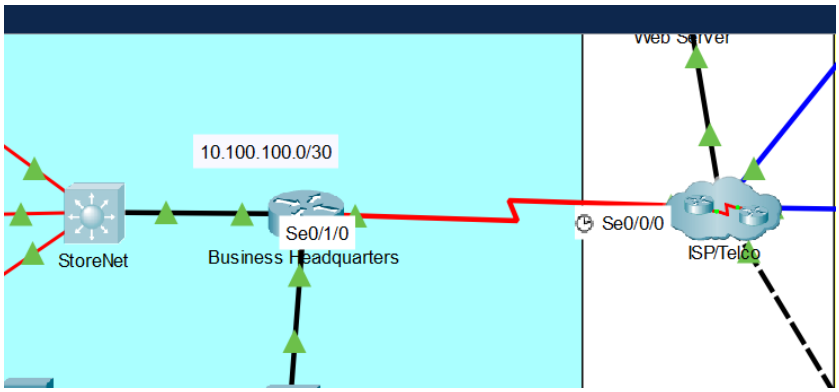


Figure 8. HQ router to ISP serial link (Se0/1/0 - Se0/0/0)

Part 2: Explore Connectivity

Ping devices within the Business WAN and the Consumer WAN networks and the web server. Can all hosts ping each other and the web server? Is this a good situation?

A host in the Business WAN (PC0, TTL 125) and a host in the Home DSL network (Laptop 1, TTL 126) both reach the web server 209.165.200.226 (3 of 4 replies; the first request times out only while ARP resolves). Full reachability between the business and consumer networks is **not a good situation**: both sit behind the same ISP with no filtering, so the business network is exposed to consumer and public traffic. Business networks should be protected with a firewall, ACLs or a VPN so that only the required services are reachable.

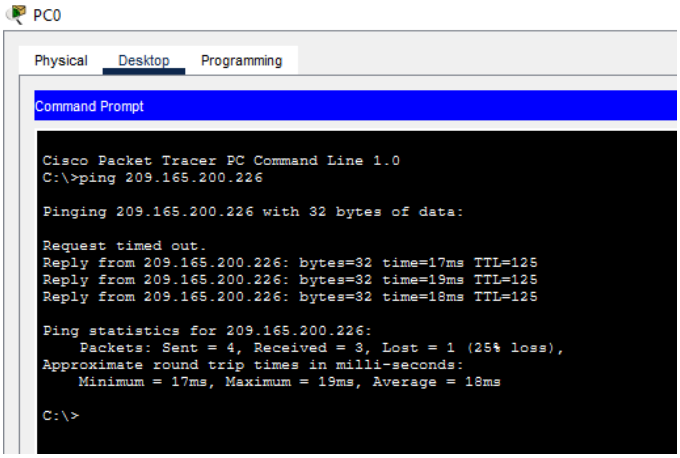


Figure 9. Ping from PC0 (Business WAN) to the web server

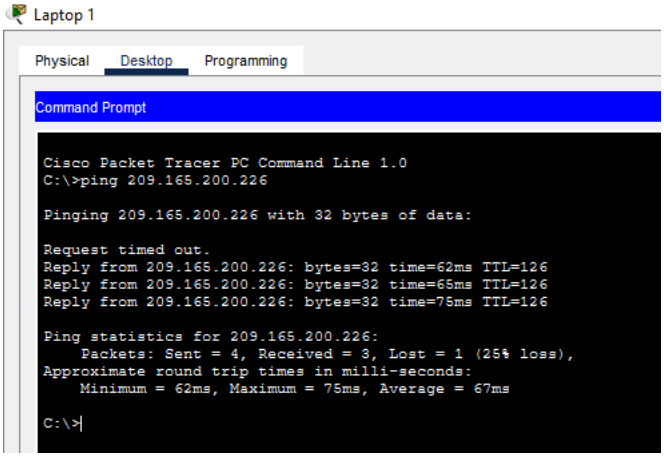


Figure 10. Ping from Laptop 1 (Home DSL) to the web server